COMPONENTS & CHIPS

Power integrated modules

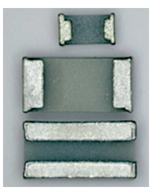
The EliteSiC power integrated modules (PIMs) are notable for their scalability, supporting output powers ranging from 25kW to 100kW. This scalability

makes them suitable for various applications, including multiple DC fast charging and energy storage system platforms. A vital feature of these systems is bidirectional charging. The company claims to offer designers the versatility to select the most suitable PIMs for the power conversion stages in applications like DC fast charging or energy storage systems.

Onsemi
https://www.onsemi.com

Surface-mount thermal jumper chips

The TMJ series features an electrically isolated jumper with thermal conductivity, maintaining capacitance and insulation resistance between its



terminals. This component uses an aluminium nitride substrate for thermal conductance, reaching up to 216mW/°C, while ensuring isolation for components that cannot be electrically grounded. Aluminium nitride (AIN) has a thermal conductivity of 170W/mK, aiding in heat dissipation and contributing to its capacitance for electrical interference. The AIN substrate also exhibits insulation resistance, ensuring performance. The chips are available in sizes ranging from 0603 to 2512, and they are designed to operate within a temperature range of -55°C to +155°C.

Stackpole Electronics
https://www.seielect.com

Infrared receiver module upgrades

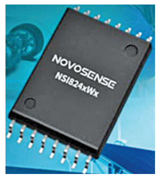
Vishay Intertechnology, Inc. has upgraded its TSOP18xx, TSOP58xx, and TSSP5xx infrared (IR) receiver module series, tailored for remote control, proximity sensing, and light barriers. Each module integrates a photodetec-

tor, preamplifier circuit, and IR filter, promoting extended battery life in portable devices and reliability outdoors. These advancements guarantee quicker lead times for customers and eliminate the need for PCB redesigns. The modules cater to diverse IR remote control applications and are optimised for various consumer electronics, with the TSSP5xx series specialising in proximity and presence sensing, detecting up to two metres, which is ideal for interactive toys, security systems, and home automation appliances.

Vishay Intertechnology, Inc.
https://www.vishay.com

Quad-channel digital isolators

NOVOSENSE Microelectronics has expanded its digital isolator portfolio with the introduction of the NSI824x-Wx quad-channel devices, tailored for industrial and automotive sectors. These high-reliability isolators, meeting UL1577 standards, combine space efficiency and advanced protection, featuring seamless level shifting for applications such as solar inverters, energy storage, industrial automation, and electric vehicles. With a wide 2.5V to 5.5V input voltage range, these isolators accommodate various configurations and offer low power



consumption of 1.5mA per channel at 1Mbps. Utilising a unique capacitive isolation technology with adaptive OOK modulation, they ensure minimal EMI and superior noise resistance. They operate across -55°C to 125°C, adhering to major safety standards and are versatile for a wide range of industrial and automotive applications.

NOVOSENSE Microelectronics
https://www.novosns.com

Ethernet switches

Microchip Technology unveils its LAN969x Ethernet switches, bolstering the industrial automation sector by integrating machine learning and robotic systems. These switches,

featuring time-sensitive networking, offer bandwidths from 46Gbps to 102Gbps, driven by a 1GHz single-core Arm Cortex-A53 CPU. Ideal for deterministic communication in industrial automation, the switches support high-availability seamless redundancy (HSR) and parallel redundancy protocols, ensuring reliability. Moreover, the switches boast enhanced security features like secure boot, firmware execution, and cryptographic libraries, ensuring robust protection against